Geometry Mediates Excitation, Resonance, and Identifiability in Fluid-Filled Viscoelastic Shells

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Abstract

The resonant behaviour of fluid-filled viscoelastic shells arises repeatedly in biomechanics, postharvest sensing, and non-destructive evaluation, yet each application is typically treated in isolation. This paper advances a unifying thesis: *shell geometry simultaneously filters the forcing field, organises the modal spectrum, and determines the invertibility of the corresponding parameter-estimation problem.*

We formalise this thesis through four results. First, we prove that equivalent-radius (spherical) reductions of non-spherical shells are *rank-deficient* in the sense of structural identifiability: geometry and material parameters become confounded, rendering independent recovery impossible. Second, we show that asphericity lifts this rank deficiency through curvature-dependent mode splitting, providing the additional spectral information needed for unique inversion. Third, we derive the near-spherical conditioning asymptotics that govern the transition from identifiable to non-identifiable regimes, classifying shells by their curvature parameter ε. Fourth, we demonstrate that forward-model adequacy does not imply inverse-model adequacy: an equivalent-sphere model may predict resonant frequencies within engineering tolerance yet fail catastrophically when used for parameter estimation.

These four results are supported by cross-application evidence drawn from the human abdomen (infrasound resonance), the watermelon (acoustic ripeness assessment), and the urinary bladder (fill-dependent resonance), demonstrating that geometry is the hidden variable unifying apparently disparate vibroacoustic phenomena.

Keywords: fluid-filled shells; viscoelastic resonance; structural identifiability; inverse problems; asphericity; Kac problem

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1 Introduction

Spectral questions recur across mathematics, physics, and engineering. Given an observed set of resonances, what can one infer about the system that produced them? Kac's celebrated formulation — "Can one hear the shape of a drum?" — remains the canonical version of this problem [7]. Its enduring influence lies not in the narrow case of planar drums, but in the larger programme it inaugurated: to understand which features of a physical system are encoded in spectral data, which are hidden by symmetry, and which are unrecoverable without additional information. The classical counterexample of isospectral but non-congruent domains [5] did not end that programme. It sharpened it. Inverse vibration theory, model updating, and elastographic parameter recovery all ask variations of the same question: when does a resonance spectrum carry enough independent information to support reliable inference about geometry, material properties, or both [3, 4, 17, 2, 1]?

In this paper we ask that question for fluid-filled viscoelastic shells. This class of systems appears in apparently disparate settings: the human abdomen and bladder in biomechanics, gas-pocket and sub-bass coupling in physiological acoustics, and fruit-ripeness assessment in postharvest sensing [10, 14, 13, 16, 11]. Yet these applications are linked by one structural fact. The shell geometry does not merely set a boundary condition for a forward calculation. It mediates the entire problem. Geometry filters how external forcing couples into admissible deformation fields; it organises the modal spectrum through curvature, symmetry, and fluid loading; and it determines whether the resulting spectrum can be inverted to recover the parameters that generated it. Our central thesis is therefore simple: *in fluid-filled viscoelastic shells, geometry does three jobs — it filters forcing, organises spectra, and determines invertibility.*

The first of these roles emerges on the excitation side. Paper 1 showed that a fluid-filled oblate shell can possess a genuine low-frequency flexural resonance near 4 Hz, yet airborne infrasound couples to that mode vanishingly weakly because the long-wavelength forcing field is itself filtered by geometry [10]. The same shell, driven mechanically through whole-body vibration, responds far more strongly because that pathway is not subject to the same spatial-harmonic suppression. Paper 6 extended the same logic below 20 Hz, where structure-borne pathways again dominate airborne ones [16]. The point is broader than the brown-note problem that motivated the programme. A resonance is not an effect in isolation. It matters how forcing is projected onto the mode shapes permitted by the geometry.

The second role is spectral organisation. Fluid-filled shells do not possess a featureless ladder of frequencies. Curvature, aspect ratio, internal loading, and wall compliance sort the admissible motions into qualitatively distinct families and shift their ordering in application-specific ways [8, 18, 6]. In the abdominal model, this distinguishes low-frequency flexural modes from the much higher breathing mode dominated by the compressibility of the entrained fluid [10]. In the bladder, changing fill volume alters both size and effective stiffness, producing a non-trivial resonance trajectory rather than simple monotonic scaling [13]. In borborygmi, constrained gas pockets introduce a different but related spectral mechanism, in which shell confinement selects the audible range of bubble oscillations [9]. In watermelon tap-tone analysis, the measurable spectral shifts used informally by growers are again organised by geometry and shell mechanics rather than by material changes alone [11]. Across domains, geometry is the quiet organiser of what the spectrum is allowed to look like.

The third role is inverse. A forward model may predict frequencies accurately enough for

classification, explanation, or design, yet still fail as an inversion model because the spectrum does not separate the parameters cleanly. That distinction emerged most clearly in the fruit and organ inverse problems of Papers 7 and 8. Equivalent-radius reductions can reproduce leading-order resonance trends remarkably well, which explains their practical appeal. But by collapsing shape information into a single scalar radius they confound geometric and material effects, and the inverse map becomes nearly singular or formally rank deficient [11, 12]. Recovering invertibility requires more than a good forward fit. It requires modal data in which different modes sample curvature differently. That requirement is geometric before it is statistical.

The capstone develops this programme around four conjectures, each of which captures one aspect of the geometry–spectrum–inversion relation. The first is a *rank-deficiency conjecture*: equivalent-radius models of non-spherical fluid-filled shells generically collapse independent geometric information and therefore produce ill-posed spectral inversion problems. The second is an *identifiability-lifting conjecture*: asphericity restores invertibility because distinct modes respond differently to distinct curvature channels. The third is a *near-spherical conditioning conjecture*: the transition from recoverable to unrecoverable inversion is controlled by regular asymptotics near the spherical limit, where symmetry suppresses the very sensitivities on which inference depends. The fourth is a *forward–inverse adequacy conjecture*: a model may be adequate for predicting resonant frequencies while remaining inadequate for parameter recovery. Taken together, these conjectures recast several apparently separate Browntone results as instances of one general statement about how geometry governs both excitation and inference.

The evidence for that statement was assembled in three acts. Act I, Papers 1–2, posed the question. The brown-note analysis established the coupling disparity between airborne and mechanical excitation, while the gas pocket study showed that alternative resonant substructures can couple far more efficiently than the whole cavity [10, 14]. Act II, Papers 3–6, broadened the evidence base: scaling laws showed that the dimensionless resonance structure persists across body sizes; bladder resonance revealed geometry–stiffening competition in another organ; borborygmi provided an audible constrained-bubble counterpart; and sub-bass coupling extended the excitation analysis to lower frequencies and alternative pathways [15, 13, 9, 16]. Act III, Papers 7–8, supplied the theoretical turn. Watermelon acoustics demonstrated that shell-resonance inversion can succeed when the geometry leaves enough spectral structure to exploit, while the Kac paper showed that asphericity can lift inverse degeneracy and that near-spherical models can be forward-useful yet inverse-fragile [11, 12]. The present paper treats those results not as a sequence of curiosities, but as evidence for a single geometry-mediated theory of excitation, resonance, and identifiability.

Our aim is therefore synthetic rather than merely archival. We do not repeat the detailed derivations of the companion papers. Instead, we extract the shared mathematical structure, state the four conjectures in a common language, and show how the examples together support a general picture of fluid-filled shells as systems in which symmetry and curvature govern both what can be excited and what can be inferred. The intended scope is wider than the original applications. The same logic should matter wherever one seeks to infer geometry or material properties from resonance data in soft shell-like structures, whether in biomechanics, non-destructive evaluation, or allied inverse problems.

The paper is organised as follows. Section 2 summarises the programme-level evidence from Papers 1–8 and identifies the common geometric structure that links them. Section 3 formulates the four conjectures precisely and relates them to the underlying shell and inverse problem theory.

Section 4 gathers the principal cross-paper results into a unified comparative account, emphasising where excitation, spectral organisation, and identifiability align or diverge. Section 5 considers the broader implications for spectral inference in fluid-filled shells and for the use of reduced models in inverse settings. Section 6 closes with the capstone claim: geometry is not a secondary detail of these systems, but the mechanism that mediates their forcing, their resonances, and the possibility of hearing the parameters that made them.

2 Background

3 Theory

4 Results

5 Discussion

6 Conclusion

References

[1] Marvin M. Doyley. Model-based elastography: A survey of approaches to the inverse elasticity problem. *Physics in Medicine and Biology*, 57(3):R35–R73, 2012.

[2] Michael I. Friswell and John E. Mottershead. *Finite Element Model Updating in Structural Dynamics.* Kluwer Academic Publishers, Dordrecht, 1995.

[3] Graham M. L. Gladwell. Inverse problems in vibration. *Applied Mechanics Reviews*, 57(6):401–422, 2004.

[4] Graham M. L. Gladwell. *Inverse Problems in Vibration*, volume 119 of *Solid Mechanics and Its Applications.* Springer, 2nd edition, 2005.

[5] Carolyn Gordon, David L. Webb, and Scott Wolpert. One cannot hear the shape of a drum. *Bulletin of the American Mathematical Society*, 27(1):134–138, 1992.

[6] Miguel C. Junger and David Feit. *Sound, Structures, and Their Interaction.* MIT Press, 2nd edition, 1986.

[7] Mark Kac. Can one hear the shape of a drum? *The American Mathematical Monthly*, 73(4):1–23, 1966.

[8] Arthur W. Leissa. *Vibration of Shells.* NASA, 1973. NASA SP-288.

[9] Jonathan Mace and Brian R. Mace. Borborygmi as constrained bubble oscillations: A shell-theoretic model of gut sounds. Companion paper (Paper 5), submitted to Journal of the Acoustical Society of America, 2026.

[10] Jonathan Mace and Brian R. Mace. The brown note revisited: Infrasonic resonance of the human abdominal cavity. Companion paper (Paper 1), submitted to Journal of Sound and Vibration, 2026.

[11] Jonathan Mace and Brian R. Mace. Can you hear the ripeness? non-destructive acoustic assessment of fruit maturity via equivalent-sphere shell resonance inversion. Companion paper (Paper 7), submitted to Postharvest Biology and Technology, 2026.

[12] Jonathan Mace and Brian R. Mace. Can you hear the shape of an organ? practical identifiability of elastic shell parameters from resonant frequencies. Companion paper (Paper 8), submitted to Inverse Problems, 2026.

[13] Jonathan Mace and Brian R. Mace. Fill-dependent resonance of the human urinary bladder: Competing geometry and stiffening in a fluid-filled viscoelastic shell. Companion paper (Paper 4), submitted to Journal of Sound and Vibration, 2026.

[14] Jonathan Mace and Brian R. Mace. Gas pocket transduction in the resonant abdomen. Companion paper (Paper 2), submitted to Journal of the Acoustical Society of America, 2026.

[15] Jonathan Mace and Brian R. Mace. Scaling laws for fluid-filled viscoelastic shell resonance. Companion paper (Paper 3), submitted to Journal of Sound and Vibration (Short Communication), 2026.

[16] Jonathan Mace and Brian R. Mace. Sub-bass coupling to the abdominal cavity: Airborne and structure-borne pathways below 20Hz. Companion paper (Paper 6), submitted to Journal of the Acoustical Society of America, 2026.

[17] John E. Mottershead and Michael I. Friswell. Model updating in structural dynamics: A survey. *Journal of Sound and Vibration*, 167(2):347–375, 1993.

[18] Werner Soedel. *Vibrations of Shells and Plates.* Marcel Dekker, 3rd edition, 2004.